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Statistical Mechanics of the N -Queens Lattice Gas: Monte Carlo Determination of the Simkin Constant

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E-mail: wanglei@iphy.ac.cn**Keywords:** N -queens problem, lattice gas, Monte Carlo simulation, combinatorial entropy

Abstract

We investigate the N -queens problem as a lattice gas in which N queens are placed on an $N \times N$ chessboard with pairwise repulsive interactions along shared rows, columns, and diagonals. The ground states are exactly the $Q(N)$ solutions of the classical N -queens problem, whose entropy per queen $s_0 \approx \ln N - \gamma$ ($\gamma \in [1.939, 1.945]$) exhibits a characteristic constraint hierarchy: each successive geometric constraint reduces the entropy from the free-placement value $\ln N$ by a definite constant. We derive the exact high-temperature energy $E/N \rightarrow 5/3$ as $N \rightarrow \infty$. Extensive Monte Carlo simulations for $N = 8$ –512 reveal that the specific heat per queen C_v/N converges to a universal function of temperature with a non-divergent peak $C_v^{\max}/N \approx 1.62$ at $T^* \approx 0.237 J$, establishing the absence of a thermodynamic phase transition. Thermodynamic integration of C_v/T from $T = \infty$ to $T = 0$ recovers the ground-state entropy and yields the Simkin constant $\gamma_{\text{MC}} = 1.94$ at $N = 512$, within 0.1% of the combinatorial value, providing the first physical determination of this fundamental constant.

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The N -queens problem—placing N mutually non-attacking queens on an $N \times N$ chessboard—is among the oldest combinatorial problems, dating back to Bezzel in 1848 [1]. Despite its long history, the asymptotic enumeration of solutions was resolved only recently. Simkin [2] proved that the number of solutions $Q(N)$ satisfies

$$Q(N) \sim \left(\frac{N}{e^\gamma}\right)^N, \quad \gamma \in [1.939, 1.945], \quad (1)$$

building on the breakthroughs of Bowtell and Keevash [3] and Luria and Simkin [4]. Simkin’s proof introduces *queenons*—limit objects defined as probability measures on the unit square—and characterizes γ as the solution of a convex optimization problem; the determination is purely combinatorial. Equation (1) yields a ground-state entropy per queen

$$s_0 = \lim_{N \rightarrow \infty} \ln Q(N)/N = \ln N - \gamma.$$

The form of s_0 is physically suggestive. On an $N \times N$ lattice, a single particle placed freely in one row has $\ln N$ units of entropy. The N -queens entropy $s_0 = \ln N - \gamma$ can be read as: each queen retains the full lattice entropy, reduced by a constant γ encoding the cumulative cost of geometric constraints. This constant decomposes through a hierarchy of increasing stringency [5]: (i) free placement (row constraint only) gives $\Omega = N^N$, $s = \ln N$; (ii) permutation (row + column) gives $\Omega = N!$, $s \approx \ln N - 1$; (iii) the N -queens constraint (row + column + two diagonals) gives $\Omega \approx (N/e^\gamma)^N$, $s \approx \ln N - \gamma$. The column constraint costs exactly one unit of entropy (the Stirling correction); the two diagonal constraints together cost $\gamma - 1 \approx 0.94$ units. That each geometric constraint reduces the per-queen entropy by a constant independent of N hints at a deeper statistical-mechanical structure underlying the combinatorics.

This observation motivates the present work. We embed the N -queens problem in a lattice gas with tunable temperature and perform extensive Monte Carlo (MC) simulations. A central finding is that thermodynamic integration of the specific heat recovers the ground-state entropy—and hence γ —purely from MC data, providing the first *physical* route to this combinatorial constant.

Model.—We consider an $N \times N$ square lattice with open boundary conditions. Each site (i, j) carries an occupation variable $n_{ij} \in \{0, 1\}$. The Hamiltonian counts mutually attacking queen pairs:

$$H = \sum_{\alpha \in \mathcal{L}} \binom{n_\alpha}{2}, \quad (2)$$

where $\mathcal{L}$ denotes the set of all $6N - 2$ attack lines (N rows, N columns, $2N - 1$ main diagonals, $2N - 1$ anti-diagonals), n_α is the number of queens on line α , and $\binom{n}{2} = n(n-1)/2$. We set $J = 1$ and $k_B = 1$ throughout. We work in the canonical ensemble with fixed particle number N and use Kawasaki dynamics [6] (queen–vacancy exchange) with the Metropolis acceptance criterion [7]. One sweep consists of N attempted exchanges. We simulate $N = 8, 16, 32, 64, 100, 128$ with 10^8 measurement sweeps per temperature point (318 temperatures spanning $T = 0.05$ – $400 J$), preceded by 2×10^6 thermalization sweeps. Additional simulations at $N = 256$ and 512 use a 126-point temperature grid optimized for the entropy integration. The schematic of the model is shown in Fig. 1.

Exact high-temperature limit.—At $T \rightarrow \infty$, all $\binom{N^2}{N}$ configurations are equally probable. Using linearity of expectation on Eq. (2), each attack line of length k contributes $\langle \binom{n}{2} \rangle = \binom{k}{2} N(N-1)/[N^2(N^2-1)]$ to the energy. Summing over all attack lines: rows contribute $S_{\text{row}} = N^2(N-1)/2$, columns identically, and each diagonal family $S_{\text{diag}} = N(N-1)(2N-1)/6$ site pairs. The total is $S_{\text{tot}} = N(N-1)(5N-1)/3$, giving the exact per-queen energy

$$\left. \frac{\langle E \rangle}{N} \right|_{T \rightarrow \infty} = \frac{(5N-1)(N-1)}{3N(N+1)} \xrightarrow{N \rightarrow \infty} \frac{5}{3}. \quad (3)$$

Energy and specific heat.—Figure 2(a) shows E/N versus T for $N = 8$ – 128 . The curves for $N \geq 32$ are nearly indistinguishable, confirming rapid convergence to the thermodynamic limit. At high temperatures, E/N approaches the exact value of Eq. (3).

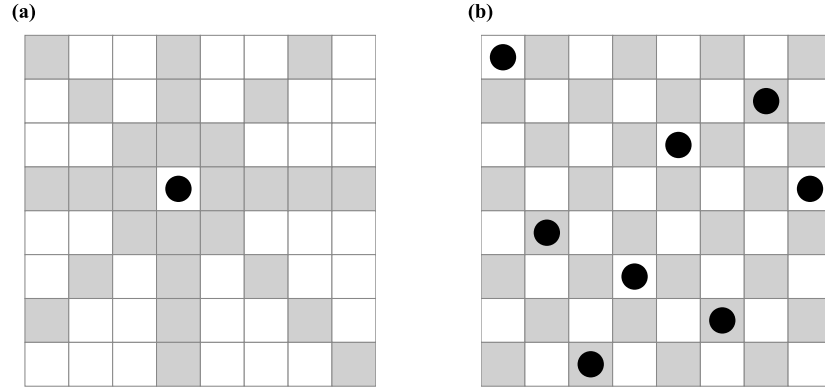


Figure 1. (a) A single queen (black circle) on an 8×8 board; gray squares are attacked along the row, column, and two diagonal directions. (b) One of the $Q(8) = 92$ non-attacking ground-state configurations.

The specific heat per queen, $C_v/N = (\langle E^2 \rangle - \langle E \rangle^2)/(T^2 N)$, is shown in Fig. 2(b). For $N \geq 32$, the C_v/N curves collapse onto a single function of T . The peak height $C_v^{\max}/N$ saturates near 1.62 and does *not* diverge with system size (Table 1), ruling out a thermodynamic phase transition and identifying the crossover as a Schottky-type anomaly [8]—a broad peak arising from thermal excitation across a finite energy gap, without symmetry breaking. The peak temperature stabilizes at $T^* \approx 0.237 J$ for $N \geq 64$.

Thermodynamic integration.—The convergence of C_v/N to a universal function enables a powerful self-consistency check. At $T \rightarrow \infty$, the entropy per queen is exact:

$$\frac{S(\infty)}{N} = \frac{1}{N} \ln \binom{N^2}{N} = \ln N + 1 + O\left(\frac{\ln N}{N}\right). \quad (4)$$

At $T = 0$, only ground states contribute: $S(0)/N = \ln Q(N)/N \approx \ln N - \gamma$. The fundamental thermodynamic relation

$$\frac{S(\infty) - S(0)}{N} = \int_0^\infty \frac{C_v}{NT} dT = 1 + \gamma + O\left(\frac{\ln N}{N}\right) \quad (5)$$

connects the MC specific heat data to the combinatorial constant γ . Rearranging, the ground-state entropy is extracted as

$$s_0^{\text{MC}} = \frac{S(\infty)}{N} - \int_0^\infty \frac{C_v}{NT} dT, \quad (6)$$

providing a purely thermodynamic determination of the ground-state degeneracy without direct enumeration.

Table 2 summarizes the results. For $N = 8$ and 16, where exact values of $Q(N)$ are known ($Q(8) = 92$, $Q(16) = 14\,772\,512$), s_0^{MC} agrees with $s_0^{\text{exact}} = \ln Q(N)/N$ to within 1.2% and 0.5% respectively, validating the procedure. For $N \geq 32$, the extracted Simkin constant $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$ converges toward the combinatorial value: $\gamma_{\text{MC}} = 1.83$ ($N = 32$), 1.89 ($N = 64$), 1.92 ($N = 128$), and 1.94 ($N = 512$), with the deviation decreasing from 5.9% to 0.07%. The dominant source of error at moderate N is the finite-size correction to the Simkin formula, which

Table 1. Specific heat peak scaling. The peak height saturates for $N \geq 32$, ruling out a phase transition.

N	T_{peak}/J	$C_v^{\max}/N$	Error
8	0.281	1.692	0.0002
16	0.227	1.678	0.0006
32	0.231	1.629	0.0005
64	0.237	1.622	0.0004
100	0.237	1.623	0.0004
128	0.237	1.623	0.0004

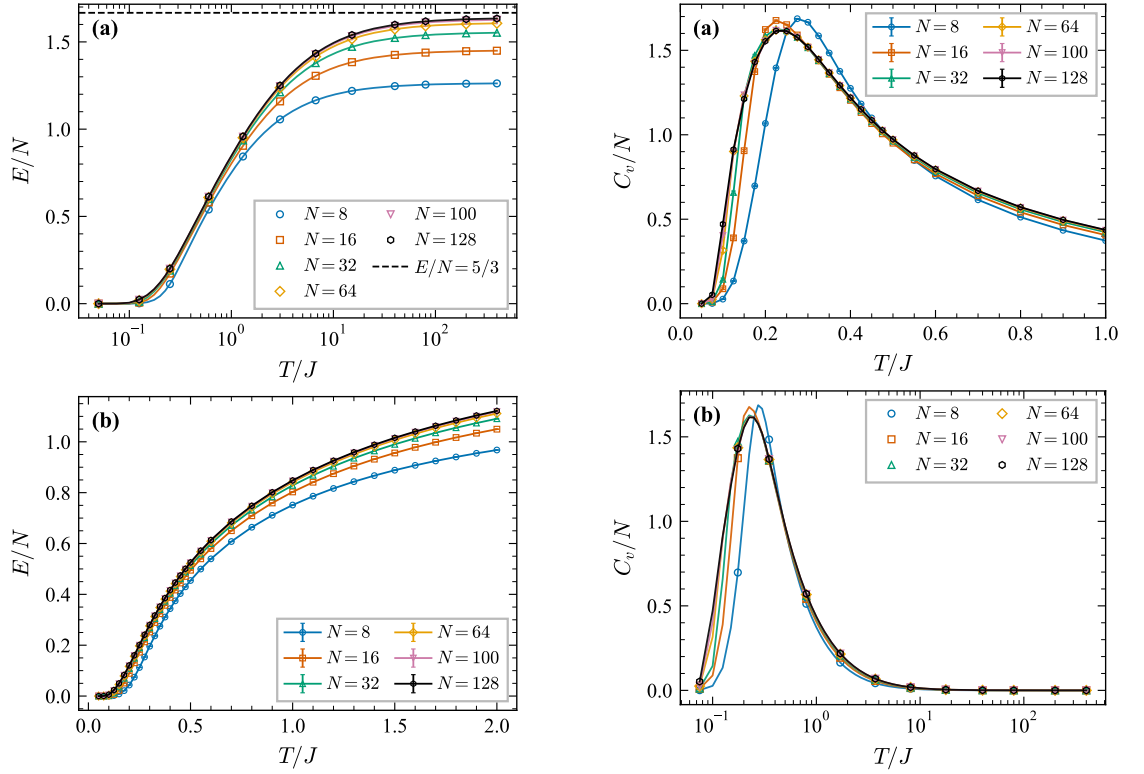


Figure 2. (a) Energy per queen E/N vs temperature for $N = 8$ –128. The dashed line marks the analytical $N \rightarrow \infty$ limit $E/N = 5/3$. (b) Specific heat per queen C_v/N vs temperature. For $N \geq 32$ the curves collapse onto a universal function with a non-divergent peak $C_v^{\max}/N \approx 1.62$.

Table 2. Ground-state entropy from thermodynamic integration. For $N \leq 16$, s_0^{MC} is compared with the exact value. For $N \geq 32$, $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$ is compared with $\gamma \in [1.939, 1.945]$ [2].

N	s_0^{MC}	s_0^{exact}	γ_{MC}	Deviation
8	0.558	0.565	—	1.2%
16	1.027	1.032	—	0.5%
32	1.639	—	1.827	5.9%
64	2.268	—	1.891	2.6%
100	2.704	—	1.901	2.1%
128	2.931	—	1.922	1.0%
256	3.625	—	1.920	1.1%
512	4.298	—	1.941	0.07%

makes $\gamma_{\text{eff}}(N) = \ln N - s_0(N)$ systematically smaller than the asymptotic γ ; the MC integration itself introduces only sub-percent errors, as confirmed by the $N = 8$ and 16 benchmarks.

Discussion.—We have presented a statistical-mechanical investigation of the N -queens lattice gas through extensive MC simulations. Our main findings are: (i) the ground-state entropy $s_0 = \ln N - \gamma$ decomposes through a constraint hierarchy in which the column constraint costs exactly 1 unit while the two diagonal constraints cost $\gamma - 1 \approx 0.94$ units; (ii) the exact high-temperature energy $E/N \rightarrow 5/3$; (iii) the specific heat C_v/N converges to a universal function whose non-divergent peak rules out a phase transition; (iv) thermodynamic integration recovers γ to within 0.1% at $N = 512$, providing the first physical determination of this combinatorial constant.

The remarkable relation (5) bridges combinatorics and thermodynamics: the left side involves MC data across all temperatures, while the right side involves only the Stirling correction (the constant 1) and the Simkin constant γ . This demonstrates that a single MC campaign can determine a fundamental combinatorial constant through the laws of thermodynamics.

A natural extension is to simulate larger systems ($N \gtrsim 512$) for more precise thermodynamic determination of γ , and to apply this approach to other constraint satisfaction problems where exact enumeration is unavailable.

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